

2. Comparison with Specialized Models

Methods	task	MAE/MSE ↓				
		Temperature	Dewpoint	Wind Direc. (×10 ³)	Wind Rate	Sea-Level Pressure
iTransformer	forecast 48h->72h	3.29/21.98	3.54/25.91	77.83/9.64	1.70/6.17	5.18/58.49
Corrformer		3/17.72	3.16/20.88	76.56/9.74	1.50/4.44	4.30/40.91
Ours		1.62/6.94	1.62/7.43	66.8/8.86	1.35/4.35	1.17/4.06
SSSD	imputation	5.49/35.21	4.51/28.23	116.82/16.46	2.52/8.85	5.67/47.54
Ours		0.91/1.80	0.90/2.12	49.0/6.01	1.00/2.09	0.56/0.61
Diffusion-TS		4.23/33.06	3.06/20.44	88.58/13.59	1.94/7.80	4.22/38.68
Ours	generation	1.79/7.27	1.82/7.93	71.9/9.50	1.43/4.47	1.27/4.57
F1-SEDI 99.5/90 ↑						
	task	Temperature	Dewpoint	Wind Direc.	Wind Rate	Sea-Level Pressure
iTransformer	forecast 48h->72h	0.15/0.52	0.1/0.45	\	0.01/0.10	0.13/0.37
Corrformer		0.17/0.56	0.28/0.51	\	0.04/0.12	0.15/0.40
Ours		0.50/0.79	0.44/0.75	\	0.24/0.43	0.76/0.86
SSSD	imputation	0.22/0.51	0.26/0.59	\	0.17/0.27	0.31/0.57
Ours		0.74/0.89	0.69/0.86	\	0.26/0.53	0.92/0.95
Diffusion-TS		0.13/0.42	0.19/0.54	\	0.15/0.25	0.21/0.49
Ours	generation	0.45/0.76	0.36/0.71	\	0.14/0.33	0.75/0.85

3. Parameter Sensitivity Analysis

UinGIO Version			MAE/MSE ↓					
	param	flops	Temperature	Dewpoint	Wind Direc. (×10^3)	Wind Rate	Sea-Level Pressure	
8 expert	3.73 MB	136.18 GFLOPs	1.58/5.82	1.58/6.33	66.98/8.75	1.33/3.82	1.12/3.53	
16 expert	4.01 MB	136.26 GFLOPs	1.59/5.86	1.58/6.30	69.32/8.73	1.33/3.85	1.10/3.49	
32 expert	4.57 MB	137.53 GFLOPs	1.53/5.61	1.56/6.25	67.72/8.69	1.33/3.78	1.04/3.24	
1M	0.99 MB	39.19 GFLOPs	1.60/5.79	1.61/6.36	66.97/8.59	1.34/3.80	1.16/3.61	
15M	14.63 MB	501.90 GFLOPs	1.50/5.39	1.54/6.09	66.31/8.65	1.31/3.74	1.01/3.02	
UinGIO Version			F1-SEDI 99.5/90 ↑					
	param	flops	Temperature	Dewpoint	Wind Direc.	Wind Rate	Sea-Level Pressure	
8 expert	3.73 MB	136.18 GFLOPs	0.54/0.80	0.46/0.75	\	0.17/0.39	0.79/0.88	
16 expert	4.01 MB	136.26 GFLOPs	0.52/0.79	0.45/0.74	\	0.18/0.39	0.78/0.88	
32 expert	4.57 MB	137.53 GFLOPs	0.55/0.80	0.46/0.75	\	0.19/0.39	0.79/0.88	
1M	0.99 MB	39.19 GFLOPs	0.52/0.79	0.45/0.74	\	0.17/0.38	0.75/0.87	
15M	14.63 MB	501.90 GFLOPs	0.55/0.80	0.47/0.75	\	0.21/0.42	0.80/0.89	
UinGIO Version			FID ↓					
	param	flops	Temperature	Dewpoint	Wind Direc.	Wind Rate	Sea-Level Pressure	
8 expert	3.73 MB	136.18 GFLOPs	0.24	0.25	0.57	0.39	0.17	
16 expert	4.01 MB	136.26 GFLOPs	0.24	0.25	0.56	0.38	0.17	
32 expert	4.57 MB	137.53 GFLOPs	0.23	0.24	0.55	0.37	0.15	
1M	0.99 MB	39.19 GFLOPs	0.25	0.27	0.58	0.44	0.19	
15M	14.63 MB	501.90 GFLOPs	0.22	0.24	0.53	0.33	0.14	
UinGIO Version			STE ↓					CroS ↓
	param	flops	Temperature	Dewpoint	Wind Direc. (×10^3)	Wind Rate	Sea-Level Pressure	
8 expert	3.73 MB	136.18 GFLOPs	2.58	1.51	2.53	1.04	0.85	1.49
16 expert	4.01 MB	136.26 GFLOPs	2.57	1.5	2.51	1.03	0.84	1.49
32 expert	4.57 MB	137.53 GFLOPs	2.55	1.49	2.47	1	0.82	1.47
1M	0.99 MB	39.19 GFLOPs	2.7	1.54	2.61	1.09	0.9	1.5
15M	14.63 MB	501.90 GFLOPs	2.5	1.45	2.36	0.97	0.76	1.44